MA 324

STATISTICAL INFERENCE AND MULTIVARIATE ANALYSIS

IIT GUWAHATI

END-SEMESTER EXAMINATION

14:00–17:00 IST

MAY 8, 2026

PART-A

Name of the Student: Harnika Dandamudi.

Signature of the Student: D. Harnika.

Roll No.: 230123020.

Signature of the Invigilator:

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

FOR OFFICE USE ONLY

Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	0	6	11

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2.

constant

$Y_1 = \theta_1 + \theta_2 + \epsilon_1$; $Y_2 = \theta_1 - 2\theta_2 + \epsilon_2$; $Y_3 = 2\theta_1 - \theta_2 + \epsilon_3$. Find $\hat{\theta}_1$ & $\hat{\theta}_2$.

$y_1 = \hat{\theta}_1 + \hat{\theta}_2 + \epsilon_1$.

$y_2 = \hat{\theta}_1 - 2\hat{\theta}_2 + \epsilon_2$.

$y_3 = 2\hat{\theta}_1 - \hat{\theta}_2 + \epsilon_3$.

We know that,

$X^T X \hat{\beta} = X^T Y$.

$\Rightarrow \hat{\beta} = (X^T X)^{-1} X^T Y$.

residual

$SS_{Res} = \sum_{i=1}^{n} (Y - X\hat{\beta})^T (Y - X\hat{\beta})$

$= \sum (Y^T - \hat{\beta}^T X^T)(Y - X\hat{\beta})$.

$0 = \sum (Y^T Y - Y^T X\hat{\beta} - \hat{\beta}^T X^T Y + \hat{\beta}^T X^T X \hat{\beta})$

$\sum (Y^T Y - 2Y^T X\hat{\beta} + \hat{\beta}^T X^T X \hat{\beta}) = 0$ — (1)

Differentiate (1) w.r.to $\hat{\beta}$.

$\Rightarrow -2Y^T X + 2\hat{\beta}(X^T X) = 0 \Rightarrow (X^T X)\hat{\beta} = Y^T X$.

$\Rightarrow (X^T X)\hat{\beta} = X^T Y$.

Now,

$\rightarrow \boxed{\hat{\beta} = (X^T X)^{-1} X^T Y}$. If $(X^T X)$ is invertible.

LSE of β: (parameters)

$n = 3$, $p = 2 \rightarrow (\theta_1, \theta_2)$

Given

$$\underset{n\times 1}{\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}} = \underset{n\times p}{\begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}} \underset{p\times 1}{\begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}} + \underset{n\times 1}{\begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{bmatrix}}.$$

Given y_i $(i = 1, 2, 3) \Rightarrow y_1, y_2, y_3$ are the observed values of the response variable.

$$\Rightarrow Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}, \quad X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}. \quad \text{Now, } X^T = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}.$$

$$X^T X = \underset{2\times 3}{\begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}} \underset{3\times 2}{\begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}} = \underset{2\times 2}{\begin{bmatrix} 1+1+4 & 1-2-2 \\ 1-2-2 & 1+4+1 \end{bmatrix}} = \underset{2\times 2}{\begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}}.$$

$(X^TX)^{-1} = \dfrac{Adj[X^TX]}{det(X^TX)}$, $det(X^TX) = 36-9 = 27 \neq 0 \Rightarrow$ Invertible. ✓.

$\left\{ \because Adj\begin{bmatrix} a & b \\ c & d \end{bmatrix} = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \right\}$. $Adj[X^TX] = \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = 3\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$.

$$(X^TX)^{-1} = \frac{3}{27}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{9}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

Now $\hat{\beta} = (X^TX)^{-1}X^TY. = \hat{\theta}$ (Here).

$$= \frac{1}{9}\underbrace{\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}}_{2\times2}\underbrace{\begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}}_{2\times3}.\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3\times1}.$$

$$= \frac{1}{9}\begin{bmatrix} 2+1 & 2-2 & 4-1 \\ 1+2 & 1-4 & 2-2 \end{bmatrix}_{2\times3}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3\times1}.$$

$$= \frac{1}{9}\underbrace{\begin{bmatrix} 3 & 0 & 3 \\ 3 & -3 & 0 \end{bmatrix}}_{2\times3}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3\times1}.$$

$$= \frac{1}{3}\underbrace{\begin{bmatrix} 1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}}_{2\times3}\begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix}_{3\times1}. \quad 2\times1.$$

$$\hat{\theta} = \frac{1}{3}\begin{bmatrix} y_1+y_3 \\ y_1-y_2 \end{bmatrix}_{2\times1} \Rightarrow \begin{bmatrix} \frac{y_1+y_3}{3} \\ \frac{y_1-y_2}{3} \end{bmatrix} = \hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$$

$\Rightarrow \therefore \hat{\theta}_1 = \dfrac{y_1+y_3}{3}$ and $\hat{\theta}_2 = \dfrac{y_1-y_2}{3}$. are the (LSE) Least Square Estimates of θ_1 and θ_2.

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2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\boldsymbol{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.
   (a) (3 points) Compute the correlation matrix $\boldsymbol{\rho}$ of $\boldsymbol{X}$.
   (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$U_1, U_2 \to$ ind. u.r.v's. on $(0,1)$. $\quad X = (X_1, X_2, X_3)'$

$X_1 = U_1$
$X_2 = U_2$
$X_3 = U_1 + U_2 \quad (X_1 + X_2)$.

(a) Correlation matrix ρ of X = ?

Variance - covariance matrix $\Sigma = \begin{bmatrix} \Sigma_{11} & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \Sigma_{22} & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \Sigma_{33} \end{bmatrix}$.

$\mathrm{Exp}(X_1) = \mu = \mathrm{Exp}(X_2)$.

$\mathrm{Exp}(X_3) = \mathrm{Exp}(X_1) + \mathrm{Exp}(X_2)$
$= 2\mu$.

$\mathrm{Var}(X_1) = \sigma^2 = \mathrm{Var}(X_2)$.

$\mathrm{Var}(X_3) = \mathrm{Var}(X_1) + \mathrm{Var}(X_2)$
$= 2\sigma^2$.

$$= \begin{bmatrix} \sigma_1^2 & \Sigma_{12} & \Sigma_{13} \\ \Sigma_{21} & \sigma_2^2 & \Sigma_{23} \\ \Sigma_{31} & \Sigma_{32} & \sigma_3^2 \end{bmatrix}.$$

Correlation matrix
$$= \begin{bmatrix} \frac{\sigma^2}{\sigma^2} & \frac{\Sigma_{12}}{\sqrt{\sigma_1^2\sigma_2^2}} & \frac{\Sigma_{13}}{\sqrt{\sigma_1^2\sigma_3^2}} \\ \frac{\Sigma_{21}}{\sqrt{\sigma_1^2\sigma_2^2}} & \frac{\sigma^2}{\sigma^2} & \frac{\Sigma_{23}}{\sqrt{\sigma_2^2\sigma_3^2}} \\ \frac{\Sigma_{31}}{\sqrt{\sigma_1^2\sigma_3^2}} & \frac{\Sigma_{32}}{\sqrt{\sigma_3^2\sigma_2^2}} & \frac{2\sigma^2}{2\sigma^2} \end{bmatrix}.$$

$2\sigma^2$

$$= \begin{bmatrix} 1 & 1 & \frac{2\sigma^2}{\sqrt{\sigma^2 2\sigma^2}} \\ 1 & 1 & \frac{2\sigma^2}{\sqrt{\sigma^2 2\sigma^2}} \\ \frac{2\sigma^2}{\sqrt{\sigma^2 2\sigma^2}} & \frac{2\sigma^2}{\sqrt{\sigma^2 2\sigma^2}} & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & \sqrt{2} \\ 1 & 1 & \sqrt{2} \\ \sqrt{2} & \sqrt{2} & 1 \end{bmatrix}$$

$\Downarrow$

Correlation matrix ρ of X.

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$\mathrm{Cov}(X_1, X_3) = \mathrm{Cov}(X_2, X_3)$.

$= E(X_1 X_3) - E(X_1)E(X_3)$. $= E[X_1(X_1 + X_2)] - E(X_1)E(X_3)$

$= E[X_1^2 + X_1 X_2] - E[X_1]E[X_3]$

$= E(X_1^2) + E(X_1^2) - E(X_1)E(X_3) =$

$= 2(\mu^2 + \sigma^2) - \mu(2\mu)$.

$= 2\mu^2 + 2\sigma^2 - 2\mu^2 = 2\sigma^2$.

$\mathrm{Var}(X_1) = E(X_1^2) - (E(X_1))^2$

$\sigma^2 = E(X_1^2) - \mu^2$.

$\mu^2 + \sigma^2 = E(X_1^2)$.

$a = u\hat{\lambda}_1$:

(b) $\rho_1^* = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

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3. Let the matrix of distance be

$$\begin{array}{c|c} & \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & & \\ 1 & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

i. Complete linkage hierarchical procedure.

ii. Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Complete linkage hierarchical procedures.

$d_{uv,w} = \max\{d_{uw}, d_{vw}\}$.

Average linkage hierarchical procedure:

$$d_{uv,w} = \frac{d_{uw} + d_{vw}}{|u||v|}.$$

(a) $(1,2) \to \underline{\min} = 1$.

$(1,2) \to$ clustered.

(i)

$$\begin{array}{c|c} & \begin{array}{ccc} (12) & 3 & 4 \end{array} \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & \\ 11 & 0 & \\ 5 & 3.5 & 0 \end{bmatrix} \end{array}$$

$d_{12,3} = \max\{d_{13}, d_{23}\}$

$= \max\{11, 2\} = 11$.

$d_{12,4} = \max\{d_{14}, d_{24}\}$

$= \max\{5, 3\} = 5$.

next $\underline{\min}$ value $= \underline{3.5}$. $d_{(3,4)}$

$(3,4) \to$ clustered.

$$\begin{array}{c|c} & \begin{array}{cc} (12) & (34) \end{array} \\ \begin{array}{c} (12) \\ (34) \end{array} & \begin{bmatrix} 0 & \\ 11 & 0 \end{bmatrix} \end{array}$$

$d_{12,34} = \max\{d_{13}, d_{14}, d_{23}, d_{24}\}$.

$= \max\{11, 5, 2, 3\} = 11$.

$\therefore \{(12), (34)\}$ clusters.

(ii) $(1,2) \to \min = 1$.

$(1,2) \to$ clustered. $\Rightarrow$

$$\begin{array}{c|c} & \begin{array}{ccc} (12) & 3 & 4 \end{array} \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & 3.5 & 0 \end{bmatrix} \end{array}$$

$$d_{12,3} = \frac{d_{13} + d_{23}}{2}$$

$$= \frac{11 + 2}{2} = \frac{13}{2} = 6.5.$$

$$d_{12,4} = \frac{d_{14} + d_{24}}{2}$$

$$= \frac{5 + 3}{2} = 4.$$

next min. value $= 3.5$ $d(3,4)$.

$(3,4) \rightarrow$ clustered.

$11+10 = \frac{21}{4}$.

$$\begin{array}{c|cc} & (12) & (34) \\ \hline (12) & 0 & \\ (34) & 5.25 & 0 \end{array}$$

$$d_{12,34} = \frac{d_{13} + d_{14} + d_{23} + d_{24}}{(2)(2)}$$

$$= \frac{11+5+2+3}{4} = 5.25$$

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$\therefore \{(12), (34)\}$ clusters

(b) Initially, $\{1\} \{2\} \{3\} \{4\}$. (i) For complete linkage hierarchical procedure.

$\downarrow$ $d_{12} = 1 \Rightarrow$ min.

$\{12\} \{3\} \{4\}$.

$\downarrow$ $d_{34} = 3.5 \Rightarrow$ min.

$\{12\} \{34\}$. $\rightarrow$ final clusters.

Dendogram ??

(ii) For Average linkage hierarchical procedure,

Initially, $\{1\} \{2\} \{3\} \{4\}$.

$\downarrow$ $d_{12} = 1 \Rightarrow$ min.

$\{12\} \{3\} \{4\}$.

$\downarrow$ $d_{34} = 3.5 \Rightarrow$ min.

$\{12\} \{34\}$. $\rightarrow$ final clusters.

0 0

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Additional Pages for Answers